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HAMBURG

15. - 17. APRIL 2026

Ship Copilot Studio Agents with Confidence

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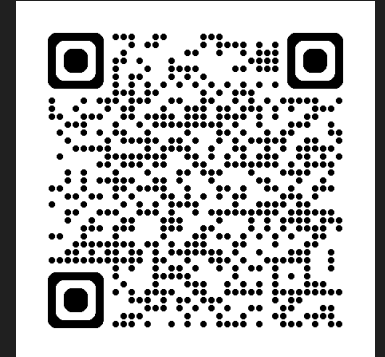




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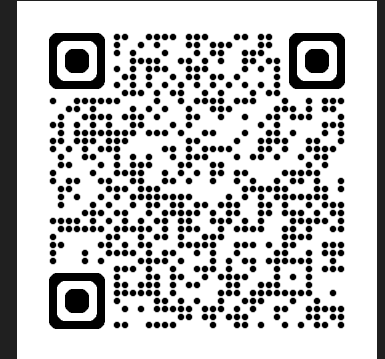


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Agenda

- **Why Automated Evaluation is Non-Optional**

Exploring the necessity of automated evaluation in modern development environments.

- **Verification vs. Validation**

Clarifying the distinction and importance of both concepts in quality assurance.

- **Tooling Overview**

A survey of popular tools and frameworks for automated testing.

- **Test Strategies**

Discussing effective approaches to designing and implementing tests.

- **Flow & Q&A**

Flow Recap & questions and discussion.

Disclaimer: All images without visible faces are generated by AI.

but why?



Remember this little guy?

**Don't let your agents leave the herd
and wander toward the mountains ...**

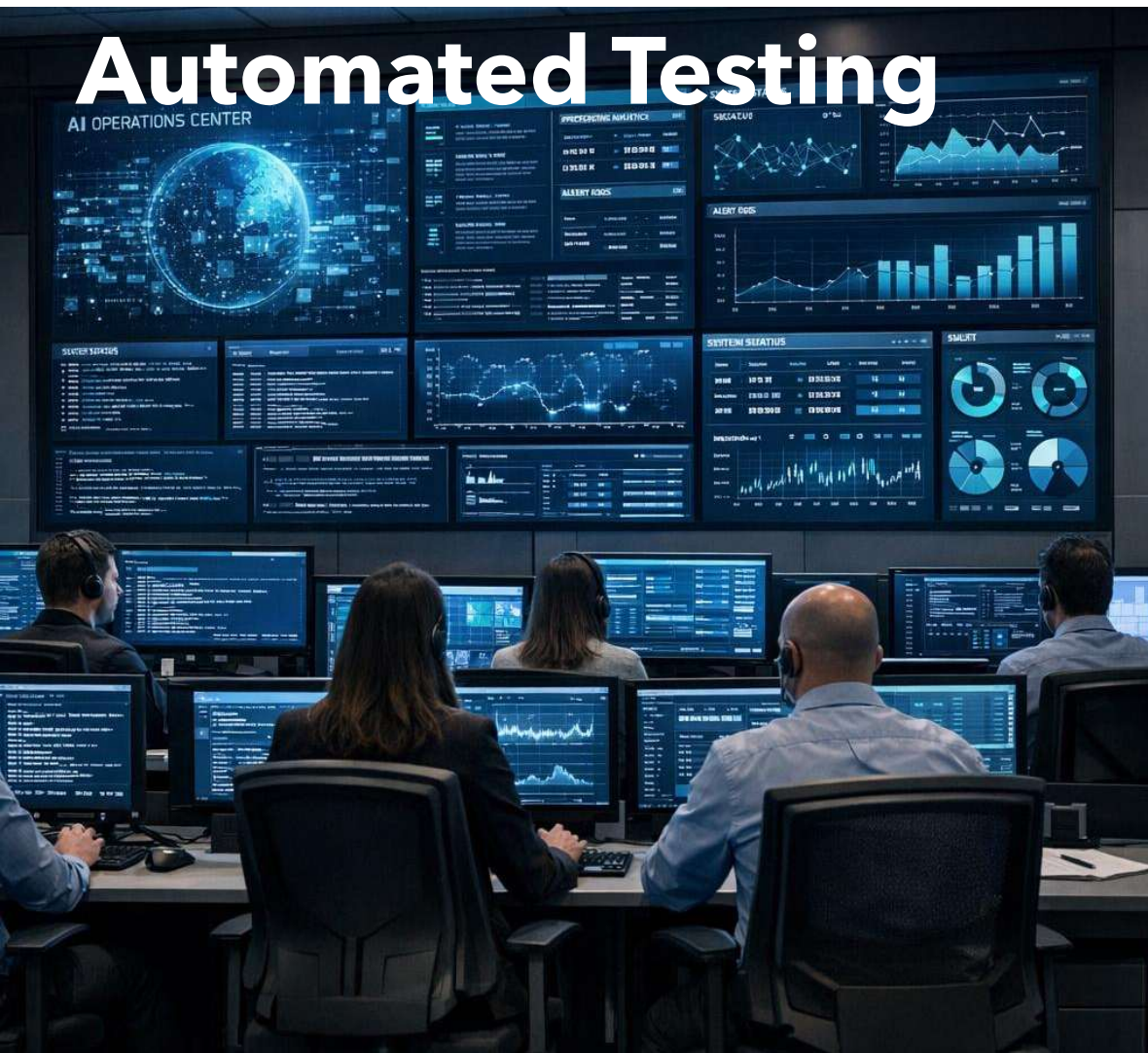
do not blame the model



Can you improve
your agents?

for sure





Automated Testing

Is Non-Optional

Casual chat tests are insufficient

Ad-hoc trials miss regressions, edge cases, and systemic drift.

LLM non-determinism and routing drift

Stochastic outputs and topic routers shift over time, requiring guardrails.

Integration fragility

APIs, schemas, and tools evolve; silent failures degrade reliability.

Need for batch, automated evaluations

Repeatable test suites at scale surface regressions and quantify impact.

Agent Evaluation & Copilot Studio Kit

Enable structured tests, batch runs, and richer diagnostics for agents.

Business outcomes

Fewer escalations, stabilized CSAT, and provable metric-based improvements.

Results that matter to business & IT

IT and architecture outcomes

Reproducible tests, audit trails, safer rollouts, and measurable improvements sprint-over-sprint.

Business outcomes

Fewer escalations, improved resolution rates and CSAT, and defensible go-lives backed by metrics.

Verification vs. Validation

Building Something Right vs. Building the Right Thing

Verification — meeting specified requirements

NASA-style: **Verification** provides objective evidence that **the system is *built right***, satisfying stated requirements through inspections, analyses, demonstrations, and tests.

Focus: specification compliance, correctness of implementation, traceability from requirements to artifacts, and acceptance criteria closure.

Stakeholder value: reduces integration risk and defects by proving each requirement is satisfied as designed, and **the map run is correct**.

Validation — meeting user and business intent

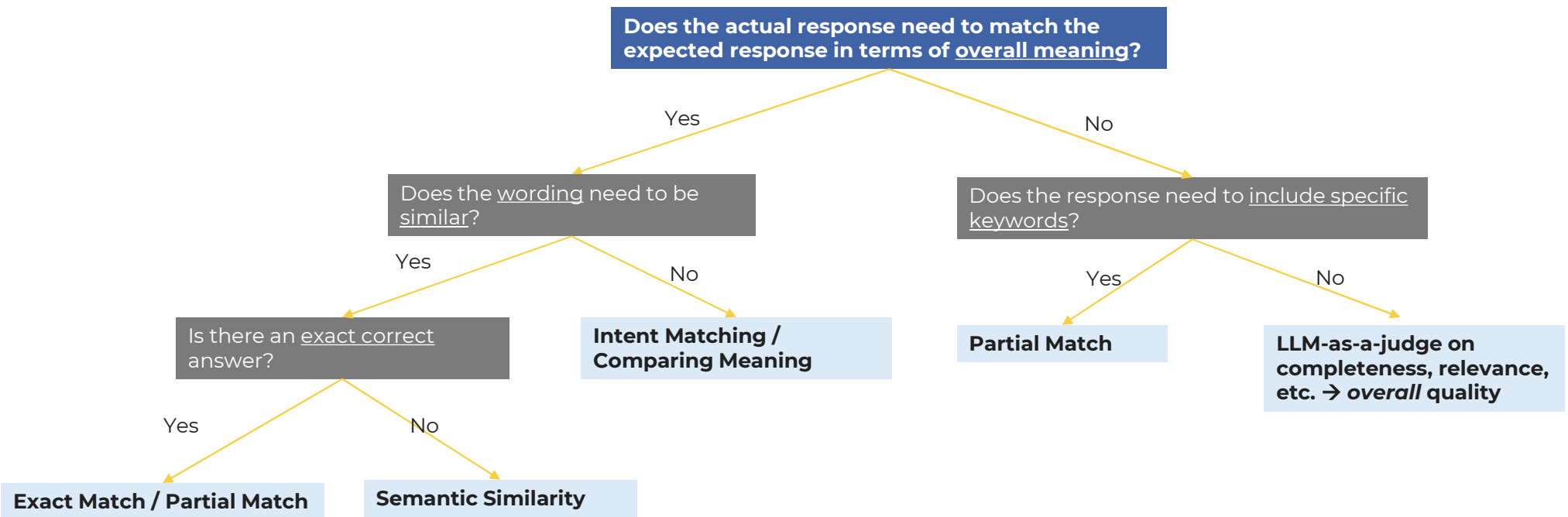
NASA-style: **Validation** provides objective evidence that **the system is the *right system***, fulfilling intended use, user needs, and mission/business objectives in its operational context.

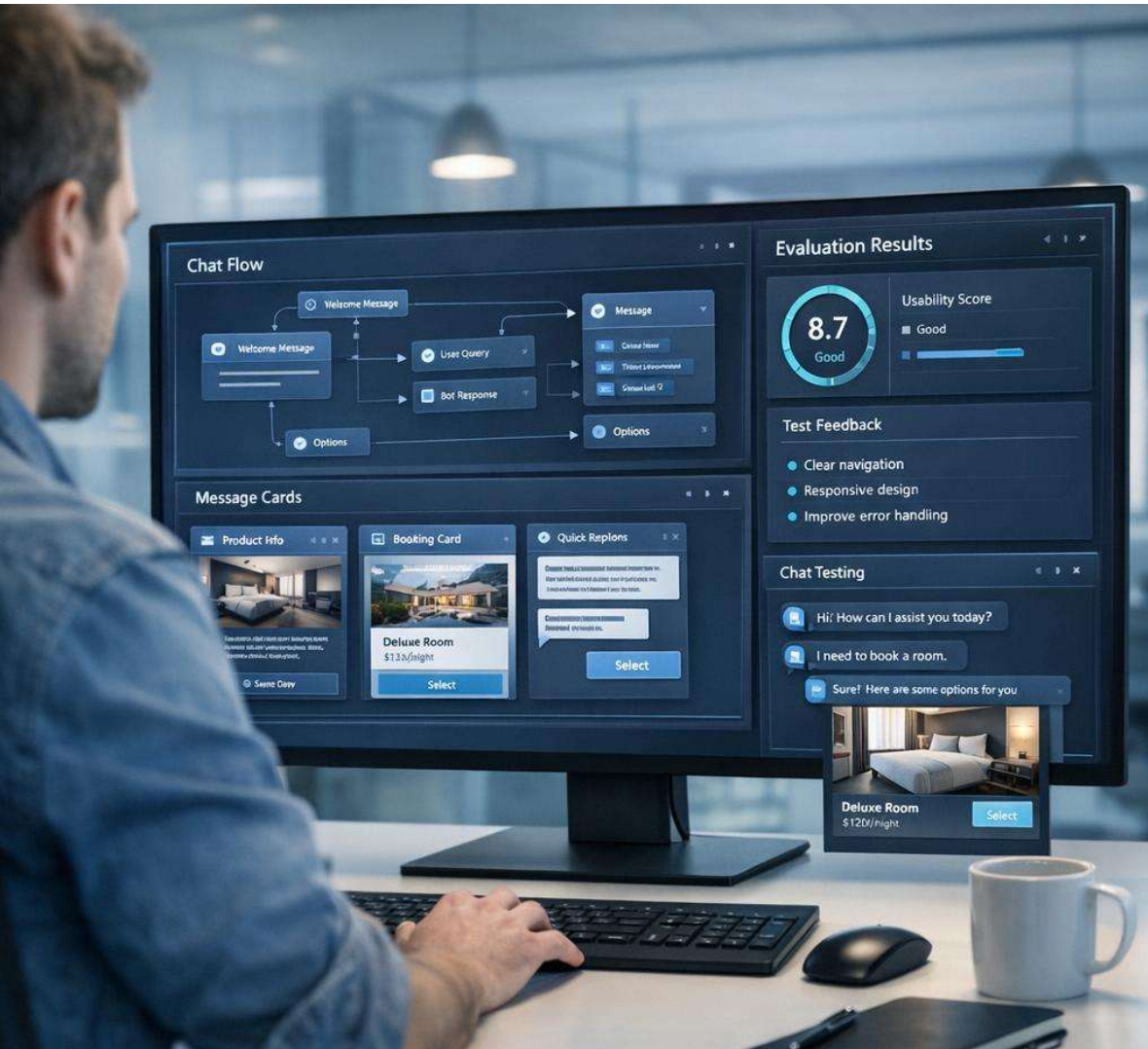
Focus: operational scenarios, end-to-end outcomes, usability, safety, and performance under realistic constraints and environments.

Stakeholder value: confirms the solution delivers expected benefits, reduces mission failure risk, and supports informed go/no-go decisions, and **the answer is as expected**.

Testing Scope for Agentic Systems

To start, ask yourself:





Agent Evaluation (in-product) – what it is

Purpose

Enable frequent, repeatable checks aligned to stakeholder goals ***directly within Copilot Studio***.

What it does

Create ***evaluation sets*** inside the studio.

Choose ***test methods*** and ***define pass/fail thresholds***.

Run ***regression checks*** to track quality over time.

Outcome

Consistent, stakeholder-aligned validation embedded in the studio workflow.

Agent Evaluation – Test Methods at a Glance

LLM-as-Judge: Quality & Intent Checks

Score relevance, groundedness, completeness, and tone; run intent recognition when supported to ensure correct routing.

Strengths: holistic, task-aware judgments.

Trade-offs: cost, latency, and potential bias; requires rubric design and periodic calibration.

Semantic Similarity: Compare Meaning

Apply embeddings or entailment to assess closeness in meaning despite phrasing variance.

Strengths: tolerant to wording and structure; suitable for summaries and paraphrases.

Trade-offs: requires similarity thresholds; may miss fine-grained errors, sensitive to domain terms.

Lexical Checks: Exact and Partial Match

Use strict equality or token overlap to verify outputs against ground truth.

Strengths: fast, deterministic, easy to automate at scale.

Trade-offs: brittle to phrasing, formatting, and legitimate variation; limited coverage for open-ended tasks.

These test methods are *complementary* tools for *different* risks.

Walkthrough

Agent Evaluation within Copilot Studio

Copilot Studio Kit:

Purpose and scope

An open-source accelerator from Power CAT that augments Copilot Studio.

Key capabilities

Batch testing, result analysis, governance dashboards (Compliance Hub), and conversation KPIs.

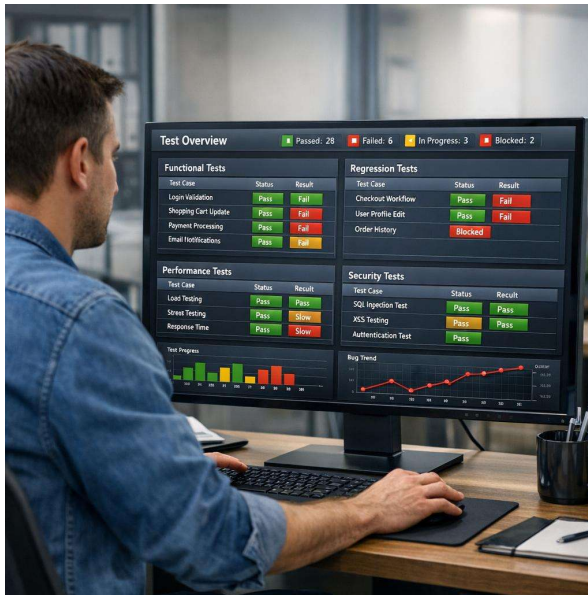
Installation options

Available via Marketplace or as a GitHub managed solution.

What It Is

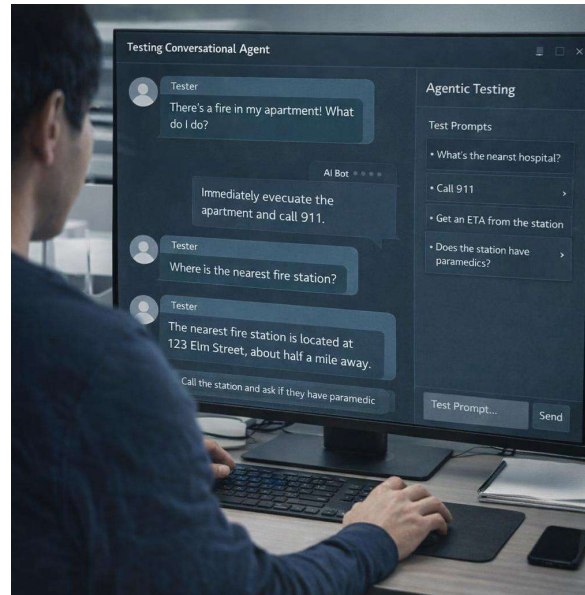


Copilot Studio Kit – Test Types



Response & Attachment Match

Use for deterministic prompts where output must **exactly match** expectations. Apply when validating strict phrasing, canonical answers, or parsing attachments reliably.



Topic Match & Generative Answers

Use topic match to verify routing to the correct domain or skill. Apply generative answer tests with **rubric-based LLM judging** via AI Builder to grade open-ended quality.



Multi-Turn Sequences & Plan Validation

Use multi-turn tests to validate end-to-end flows across steps. Apply plan validation to confirm correct tool and agent selection under generative orchestration.

Walkthrough

Copilot Studio Studio Kit

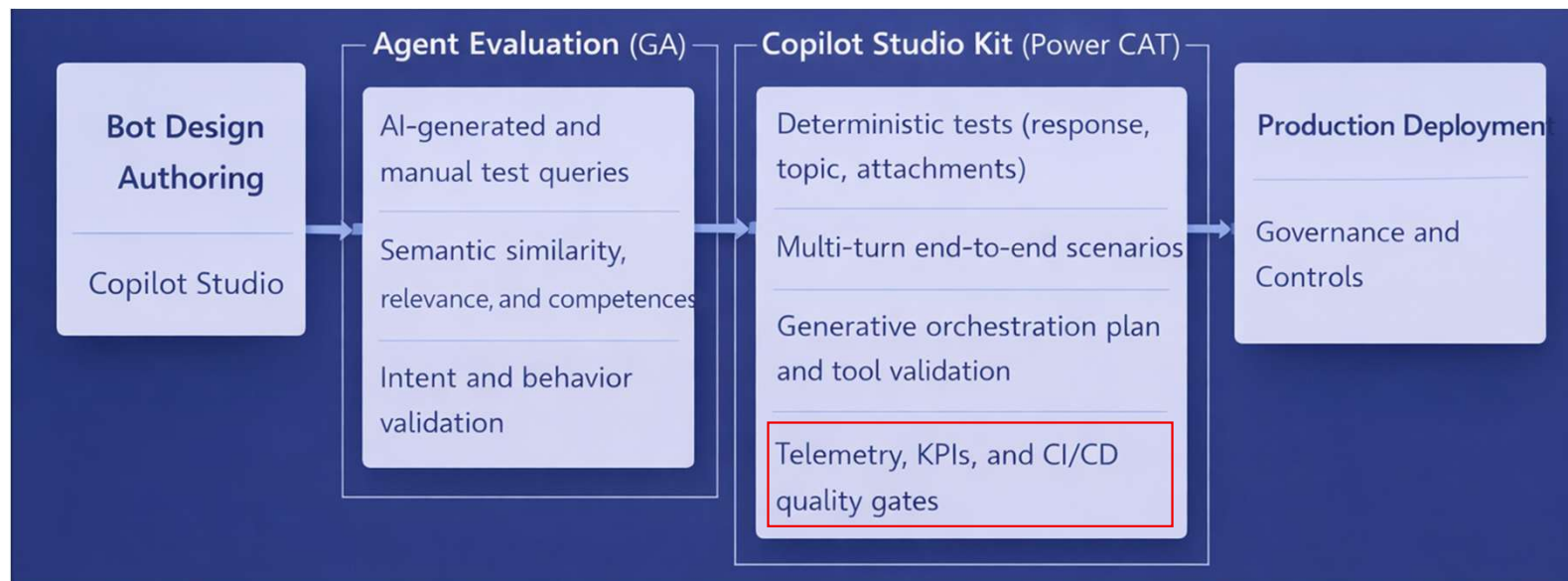
Agent Evaluation vs. Copilot Studio Kit – when to use what

Agent Evaluation — Use When

Ideal for in-studio, rapid regression loops driven by makers. **Supports threshold-based go/no-go decisions during iterative development.** Best for quick validation of incremental changes before release.

Copilot Studio Kit — Use When

Designed for scaled batch runs and multi-turn tests. **Enables plan validation and analytics enrichment via App Insights and Dataverse.** Provides governance dashboards for oversight and reporting.



These tools are complementary.

Side-by-Side Comparison

Dimension	Agent Evaluation	Copilot Studio Kit
<u>Workspace</u>	Built directly into Copilot Studio authoring UI	Separate model-driven app (Dataverse) you install
<u>Test data authoring</u>	Create/import CSV ; reuse Test Pane conversations; Manual or AI-generated prompts	Excel import/export; Manual or AI-generated prompts
<u>Execution</u>	Run directly in Copilot Studio	Via Direct Line API with cloud flows/enrichment
<u>Test methods</u>	Exact/Partial match; Similarity/Intent ; AI quality graders	Response/Attachment/Topic match; Generative Answers ; Multi-turn ; Plan validation
<u>Observability</u>	Aggregate pass/fail and scores with drill-downs in Copilot Studio	Enrichment with Dataverse storage + App Insights (topics, intent scores, latencies)
<u>CI/CD fit</u>	Strong for inner loop ; roadmap for broader scenarios, API ready	Strong for release gates , duplication, and re-runs
<u>Adjacent features</u>	Focused on evaluation inside Copilot Studio	Part of a broader kit (Compliance Hub, KPIs, SharePoint sync)



Usage in pipelines

Agent Evaluation

 HTTP

 ...

URI *

https://api.test.powerplatform.com/copilotstudio/environments/Default-c2983f0e-34ee-4b43-8abc-c2f460fd26be/bots  parentbotid (Val... X
/api/makerevaluation/testsets  TestSet X /run?api-version=1

Method *

POST 

Headers

Enter key

Enter value



Queries

Enter key

Enter value



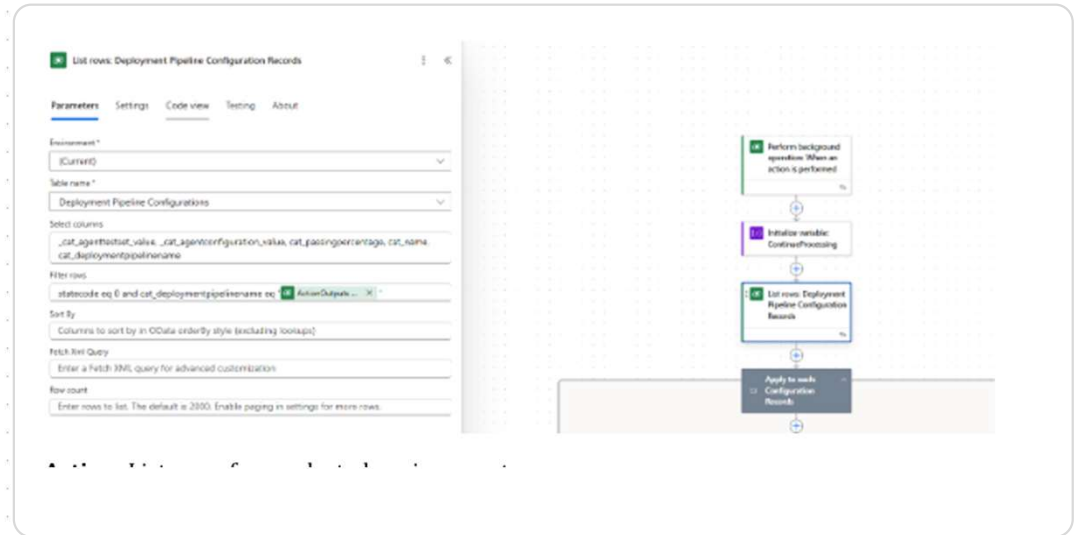
Body

```
{  
  "mcsConnectionId": "c25c8519520843a6ad35f1b78b7b167b",  
  "EvaluationRunName": "Automated Regression Test For HR Policy Agent  
   formatDate(...) X "  
}
```

Cookie

[link to documentation](#)

Copilot Studio Kit



[link to documentation](#)



Designing strong evaluation sets

Core patterns

Include not only golden paths for must-win flows, but also known-bad and negative tests, edge cases, periodicity-sensitive questions, and multilingual variants where applicable.

High-quality sources

Leverage recent test pane interactions, domain knowledge sources, and structured CSV imports to reflect real user behavior and coverage.

Curation approach

Combine AI-generated test questions with manual review to enforce correctness, consistency, and business relevance across scenarios.

**Failing tests are not a problem;
they present a great opportunity for improvement.**



From tests to quality gates in pipelines

01

Pause Deployment

Deployment is temporarily halted at a quality gate to ensure only validated code progresses.

02

Run Test Suites

Copilot Studio Kit test suites execute automatically in the pipeline host environment.

03

Parse Results

Test pass-rates and thresholds are parsed to determine release readiness across Dev, Pipeline Host, and Target environments.

04

Proceed or Block

If results meet criteria, deployment proceeds; otherwise, the release is blocked for review before targeting the destination environment.

05

Auditable Releases

Ensures all releases are documented and reduces regressions by enforcing automated quality checks.

Instrumentation and Diagnostics

Connect telemetry sources

Integrate Application Insights for runtime traces and Dataverse for structured session records. Link conversation IDs to telemetry to unify prompts, responses, and outcomes.

Capture diagnostic signals

Record topics triggered, intent scores, grounding outcomes, retrieval coverage, and failure reasons for generative answers. Persist key metrics and timestamps for drill-down analysis.

Guide iterative improvements

Use dashboards to prioritize prompt tuning, refine topic design, and address failure modes. Track changes over time to validate quality gains and operational reliability.





Multi-Agent Testing Considerations

Plan Validation in Orchestration

Verify selected tools, actions, and connected agents meet threshold criteria before execution.

Multi-Turn Test Sets

Run end-to-end scenarios across orchestrated agents to confirm hand-offs and outcomes.

Quality Tracking and Selection

Measure quality by agent model or runtime policy to choose the best configuration.

Conclusion

Use **Agent Evaluation** to *build* with confidence.
Use the **Copilot Studio Kit** to *run* with confidence.

Fewer escalations

Automated, repeatable tests reduce defects before release and lower support burden.

Improved customer satisfaction

Higher quality Copilot Studio agents deliver consistent outcomes users can trust.



Safer rollouts

Begin with in-studio evaluation and scale to enterprise testing and governance.

Measurable improvements

Auditable quality assurance with Copilot Studio Kit and Agent Evaluation accelerates continuous gains.

Join the “wildlife” care squad – today!





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THANK YOU!



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Q&A



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